Power BI Dashboard

Technical Documentation

UK Railway Analytics Dashboard

DEPI Graduation Project | Data Analysis & Power BI Track

Contributors

DEPI Graduation Project – Data Analysis & Power BI Specialist Track

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Project Overview

This project is an interactive UK Railway Analytics Dashboard built using Power BI. It provides insights into ticket sales performance, journey punctuality, delay patterns, and refund behavior using CSV-based datasets. The dashboard includes KPIs such as on-time rate, total revenue, delay causes, and refund rate to help railway operations teams make data-driven decisions.

Features

- Ticket Sales Analysis: Track revenue and volume across routes, stations, ticket types, and payment methods.
- Journey Performance Insights: Identify on-time, delayed, and cancelled journey patterns over time.
- Delay Cause Breakdown: Analyze the root causes of delays including Weather, Signal Failure, Technical Issues, Staffing, and Traffic.
- Refund Request Overview: Examine refund rates and their relationship to journey disruptions and delay reasons.
- Interactive Dashboards: Filter and visualize key metrics dynamically using slicers and cross-filtering across all pages.

1. Overview

Field	Details
Dashboard Name	UK Railway Analytics Dashboard
Purpose	Analyze railway ticket sales, journey performance, delays, and refund patterns
Primary Data Subject	Train journeys, ticket purchases, delay causes, and refund requests
Data Source	railway.csv — 31,653 transaction records
Tool	Microsoft Power BI Desktop
Track	Data Analysis & Power BI Specialist – DEPI Graduation Project

2. Data Sources

- All data loaded via a single CSV file: railway.csv
- Power Query used to clean, transform, and prepare data for modeling
- Total Records: 31,653 transactions
- Date Range: December 2023 – April 2024

2.1 Dataset Column Reference

Column Name	Description	Data Type
Transaction ID	Unique identifier for each ticket purchase	Text
Date of Purchase	Date the ticket was purchased	Date
Time of Purchase	Time the ticket was purchased	Time
Purchase Type	Online or Station purchase	Text (2 values)
Payment Method	Credit Card, Contactless, Debit Card	Text (3 values)
Railcard	Railcard type: Adult, Disabled, Senior	Text (3 values)
Ticket Class	Standard or First Class	Text (2 values)
Ticket Type	Advance, Off-Peak, Anytime	Text (3 values)
Price	Ticket price in GBP (£1 – £267)	Integer
Departure Station	Origin station (12 unique stations)	Text
Arrival Destination	Destination station (32 unique destinations)	Text
Date of Journey	Date the journey took place	Date
Departure Time	Scheduled departure time	Time
Arrival Time	Scheduled arrival time	Time
Actual Arrival Time	Actual arrival time (may differ if delayed)	Time

Column Name	Description	Data Type
Journey Status	On Time, Delayed, or Cancelled	Text (3 values)
Reason for Delay	Cause of delay when applicable	Text (8 categories)
Refund Request	Whether a refund was requested: Yes/No	Text (2 values)

3. Data Model

The data model follows a Star Schema design. One central Fact Table holds all transactional records, connected to multiple Dimension Tables for clean filtering and aggregation.

Fact Table

- railway: The main transaction table derived from railway.csv, containing all 31,653 journey records with ticket, pricing, journey status, and refund data.

Dimension Tables

Table	Key Column	Description
railway	Transaction ID	Main fact table — 31,653 journey records from railway.csv
DimPaymentMethod	Payment Method	DISTINCT(railway[Payment Method]) — Credit Card, Contactless, Debit Card
DimRailcard	Railcard	DISTINCT(railway[Railcard]) — Adult, Disabled, Senior, None
DimTicketType	Ticket Type	DISTINCT(railway[Ticket Type]) — Advance, Off-Peak, Anytime
Date	Date	Date dimension — Day, Month Name, Month Number, Month-Year, Quarter, Weekday, Year
Measures (2)	—	Dedicated table holding all DAX calculated measures

4. Transformations Summary

- Type conversions: Date of Purchase, Date of Journey cast to Date; Time columns cast to Time
- Delay Reason standardization: merged 'Signal Failure' / 'Signal failure' and 'Weather' / 'Weather Conditions' / 'Staff Shortage' / 'Staffing' into unified categories
- Created calculated column: Journey Duration (minutes) = Actual Arrival Time – Departure Time
- Created calculated column: Delay Minutes = Actual Arrival Time – Arrival Time (when Delayed)
- Created calculated column: Route = Departure Station & ' → ' & Arrival Destination
- Created calculated column: Purchase Lead Days = Date of Journey – Date of Purchase
- Created Date Dimension table using CALENDAR() in DAX for full time intelligence support
- Filtered out null/blank Reason for Delay rows when Journey Status = 'On Time' to clean visuals

5. Page Descriptions

Overview

Metrics (KPI Cards):

- Total Tickets Sold: COUNT(railway[Transaction ID])
- Total Revenue: SUM(railway[Price])
- Average Ticket Price: AVERAGE(railway[Price])
- On-Time Rate: DIVIDE(COUNTROWS(FILTER(railway, railway[Journey Status] = "On Time")), [Total Tickets])
- Delay Rate: DIVIDE(COUNTROWS(FILTER(railway, railway[Journey Status] = "Delayed")), [Total Tickets])
- Cancellation Rate: DIVIDE(COUNTROWS(FILTER(railway, railway[Journey Status] = "Cancelled")), [Total Tickets])

Visuals:

- Bar chart: Tickets Sold by Departure Station
- Line chart: Monthly Revenue Trend
- Donut chart: Journey Status Distribution (On Time / Delayed / Cancelled)
- KPI cards for all 6 metrics above

Sales & Revenue Analysis

Metrics:

- Revenue by Ticket Class: Standard vs. First Class
- Revenue by Purchase Type: Online vs. Station
- Revenue by Payment Method

Visuals:

- Stacked bar chart: Revenue by Ticket Type (Advance / Off-Peak / Anytime)
- Pie chart: Purchase Type split (Online 58.5% vs. Station 41.5%)
- Bar chart: Top 5 Busiest Routes by ticket volume
- Matrix table: Departure Station × Ticket Class with Revenue

Journey Performance & Delay

Metrics:

- Delay Rate by Route
- Average Delay Minutes
- Cancellation Rate by Station

Visuals:

- Bar chart: Delay Reasons Distribution (Weather, Technical Issue, Signal Failure, Staffing, Traffic)
- Line chart: Journey Status Over Time (monthly On Time vs. Delayed vs. Cancelled)
- Map visual: Delay rate by Departure Station
- Table: Top delayed routes with average delay minutes

Refund & Customer Behavior

Metrics:

- Refund Rate: DIVIDE(COUNTROWS(FILTER(railway, railway[Refund Request] = "Yes")), [Total Tickets])
- Refund Rate by Journey Status
- Refund Rate by Railcard Type

Visuals:

- Bar chart: Refund Requests by Reason for Delay
- Donut chart: Refund vs. No Refund split
- Line chart: Refund Requests Over Time
- Table: Passengers with Refund = Yes and their journey details

Route & Station Intelligence

Metrics:

- Busiest Route by volume
- Highest Revenue Route
- Most Delayed Route

Visuals:

- Bar chart: Top 10 Routes by Ticket Sales
- Scatter plot: Route Volume vs. Average Ticket Price
- Map: Passenger flow between stations (bubble size = volume)
- Table: Station-level KPIs — Total Tickets, Revenue, Delay Rate, Refund Rate

6. Key DAX Measures

Measure Name	DAX Formula
% Credit Card	DIVIDE([Credit Card Payments], COUNTROWS(railway))
% Online Purchases	DIVIDE([Online Purchases], COUNTROWS(railway))
% Railcard Holders	DIVIDE([Railcard Holders], COUNTROWS(railway))
% Revenue First Class	DIVIDE([First Class Revenue], [Total Revenue])
Advance Ticket Revenue	CALCULATE(SUM(railway[Price]), railway[Ticket Type] = "Advance")
Cancellation Rate	DIVIDE(CALCULATE(COUNTROWS(railway), railway[Journey Status]="Cancelled"), [Total Journeys], 0)
Cancelled Journeys	CALCULATE(COUNTROWS(railway), railway[Journey Status] = "Cancelled")
Credit Card Payments	CALCULATE(COUNTROWS(railway), railway[Payment Method] = "Credit Card")
Delayed Journeys	CALCULATE(COUNTROWS(railway), railway[Journey Status] = "Delayed")
First Class Revenue	CALCULATE(SUM(railway[Price]), railway[Ticket Class] = "First Class")
Journeys by Route	COUNTROWS(railway)
Lowest Month Revenue	CALCULATE(MINX(VALUES('Date'[Month-Year]), [Monthly Revenue]))
On-Time Journeys	CALCULATE(COUNTROWS(railway), railway[Journey Status] = "On Time")
On-Time Rate	DIVIDE([On-Time Journeys], [Total Journeys], 0)
Online Purchases	CALCULATE(COUNTROWS(railway), railway[Purchase Type] = "Online")
Railcard Holders	CALCULATE(COUNTROWS(railway), railway[Railcard] <> "None")
Standard Class Revenue	CALCULATE(SUM(railway[Price]), railway[Ticket Class] = "Standard")
Total Journeys	COUNTROWS(railway)
Total Passengers	COUNTROWS(railway)
Total Revenue	SUM(railway[Price])

7. Key Performance Indicators (KPIs)

KPI	Value (from dataset)
Total Transactions	31,653
On-Time Rate	86.8% (27,481 journeys)
Delay Rate	7.2% (2,292 journeys)
Cancellation Rate	5.9% (1,880 journeys)
Average Ticket Price	£23.44
Most Common Payment	Credit Card (60.5%)
Most Common Ticket Type	Advance (55.5%)
Top Departure Station	Manchester Piccadilly (5,650 tickets)
Busiest Route	Manchester Piccadilly → Liverpool Lime Street (4,628 journeys)
Top Delay Cause	Weather (27.9% of delays)
Refund Request Rate	3.5% (1,118 requests)

8. Project Planning & Management

Objective

Build an interactive Power BI dashboard to analyze UK railway ticket sales, journey punctuality, delay patterns, and refund behavior to support data-driven decisions for railway operations.

Scope

- Analyze ticket sales across different routes, stations, ticket types, and payment methods
- Evaluate journey performance including on-time rate, delay causes, and cancellations
- Examine refund patterns and their relationship to journey disruptions
- Include KPIs such as total revenue, delay rate, cancellation rate, and refund rate

Project Timeline

Milestone	Task	Start Date	End Date
Data Preparation	Collect and clean railway dataset	March 1, 2025	March 7, 2025
Dashboard Design	Define KPIs and page layout	March 8, 2025	March 14, 2025
Data Modeling	Transform data using Power Query, build star schema	March 15, 2025	March 21, 2025
Visualization Development	Build all 5 dashboard pages in Power BI	March 22, 2025	March 28, 2025
Interactivity & Refinement	Add slicers, filters, cross-filtering, tooltips	March 29, 2025	April 4, 2025
Final Review & Documentation	Test, validate insights, and finalize report	April 5, 2025	April 10, 2025

9. Risk Assessment & Mitigation

Risk	Impact	Solution
Inconsistent delay reason labels (e.g. 'Signal Failure' vs. 'Signal failure')	High	Text standardization in Power Query using Text.Proper() and Replace Value
Missing Reason for Delay for On Time journeys	Low	Filter nulls only in delay-specific visuals; keep all rows in the fact table
Dashboard performance with 31K+ rows	Medium	Optimize DAX with CALCULATE instead of FILTER where possible; use aggregations
Time calculations across midnight (e.g. late-night trains)	Medium	Use Date + Time combined columns; handle midnight crossover logic in Power Query
Misleading delay averages due to cancelled journeys	Medium	Apply FILTER to include only Delayed rows when calculating average delay minutes

10. System Design

Problem Statement

Railway operations teams and analysts lack a unified, interactive view of ticket sales performance, journey punctuality, and customer refund patterns. This dashboard centralizes all operational metrics into one Power BI solution.

Technology Stack

Component	Technology
Data Visualization	Microsoft Power BI Desktop
Data Processing & Cleaning	Power Query (M Language)
Business Logic & Calculations	DAX (Data Analysis Expressions)
Data Source	CSV File (railway.csv)
Deployment	Power BI Service (Cloud Sharing)
Version Control	GitHub

Data Flow

- Step 1: Raw CSV data loaded into Power Query
- Step 2: Data cleaned, columns typed, delay reasons standardized
- Step 3: Dimension tables created; star schema relationships defined
- Step 4: DAX measures calculated on top of the model
- Step 5: Visuals built on each dashboard page with cross-filter interactions
- Step 6: Published to Power BI Service for team access